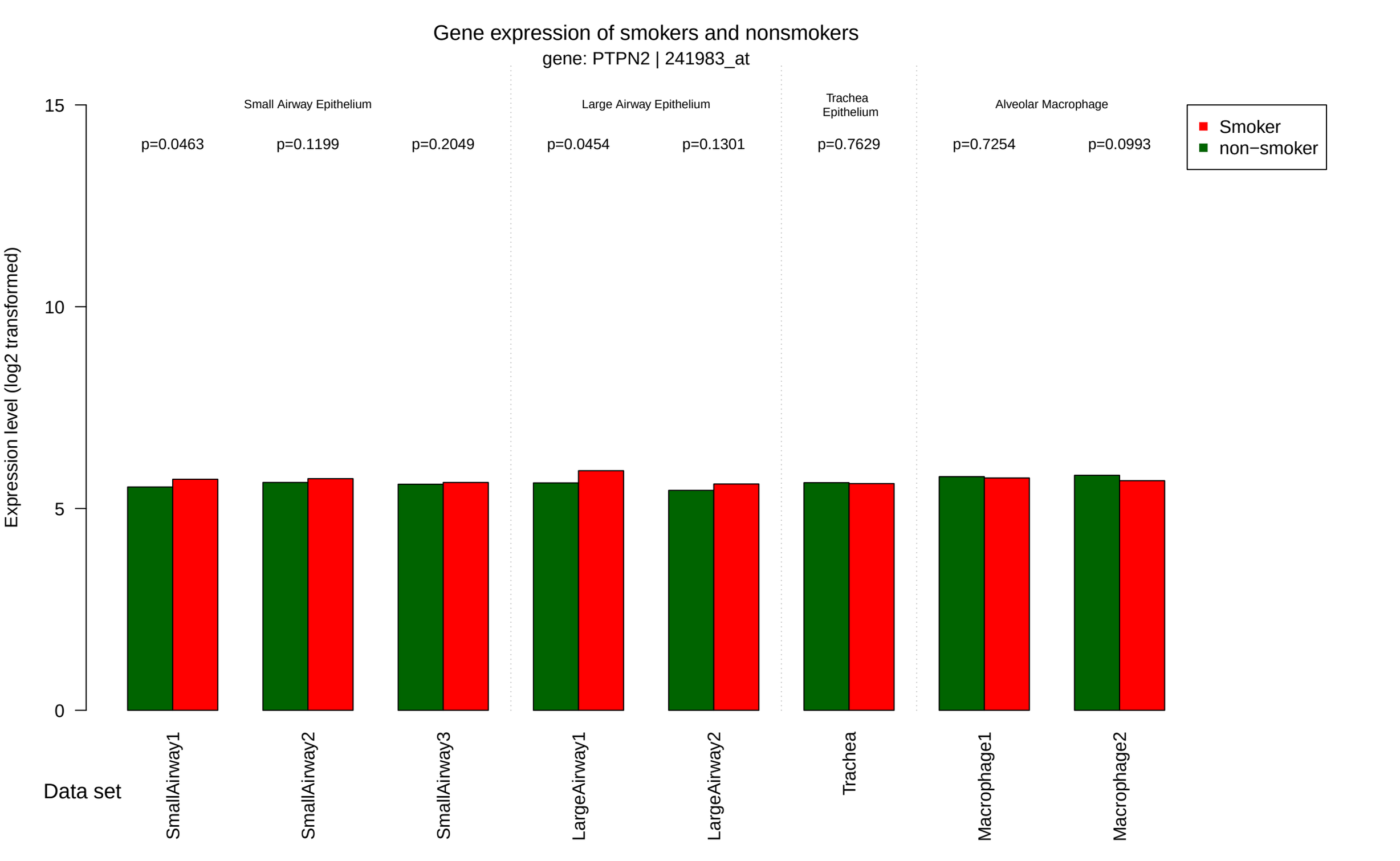
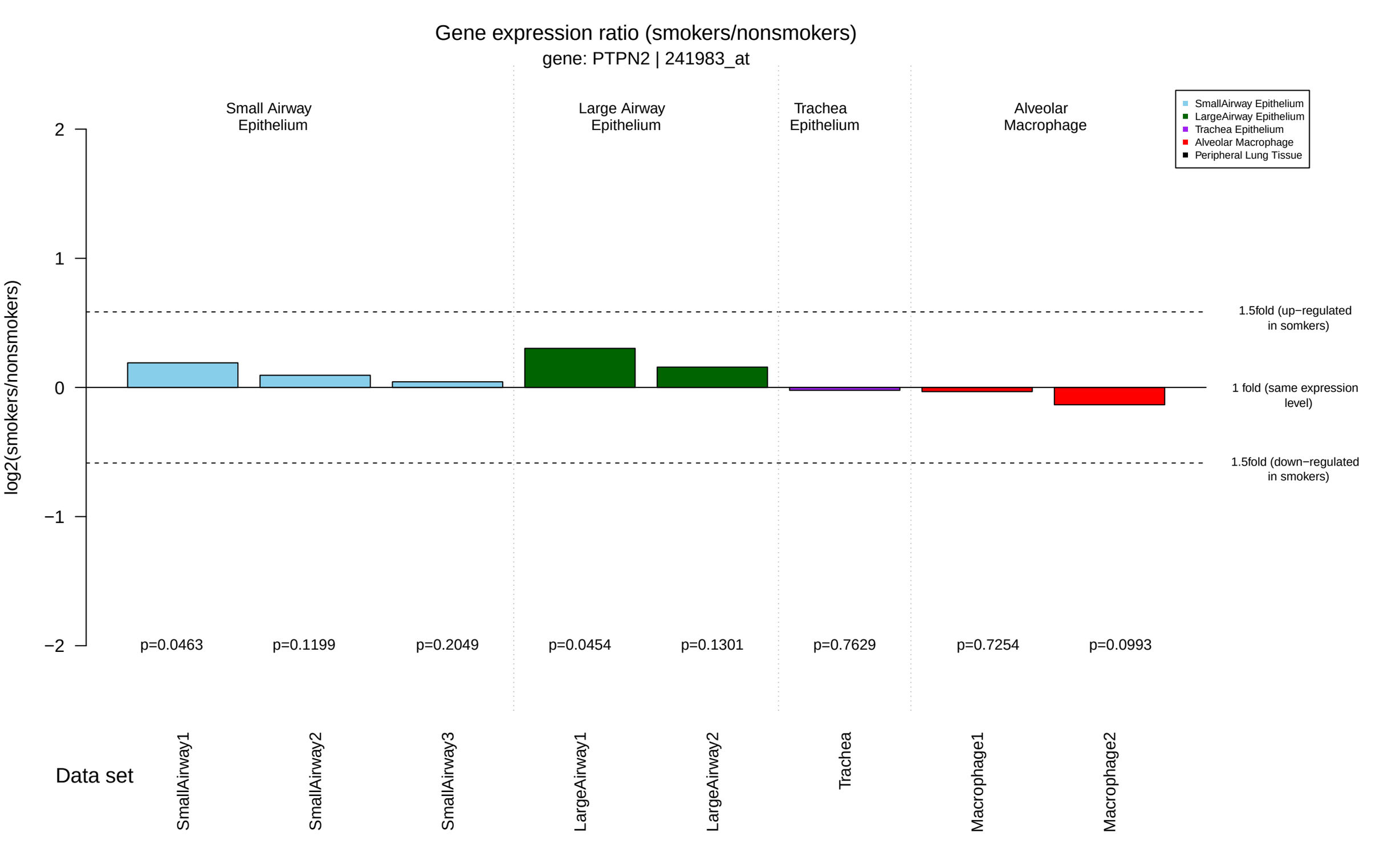
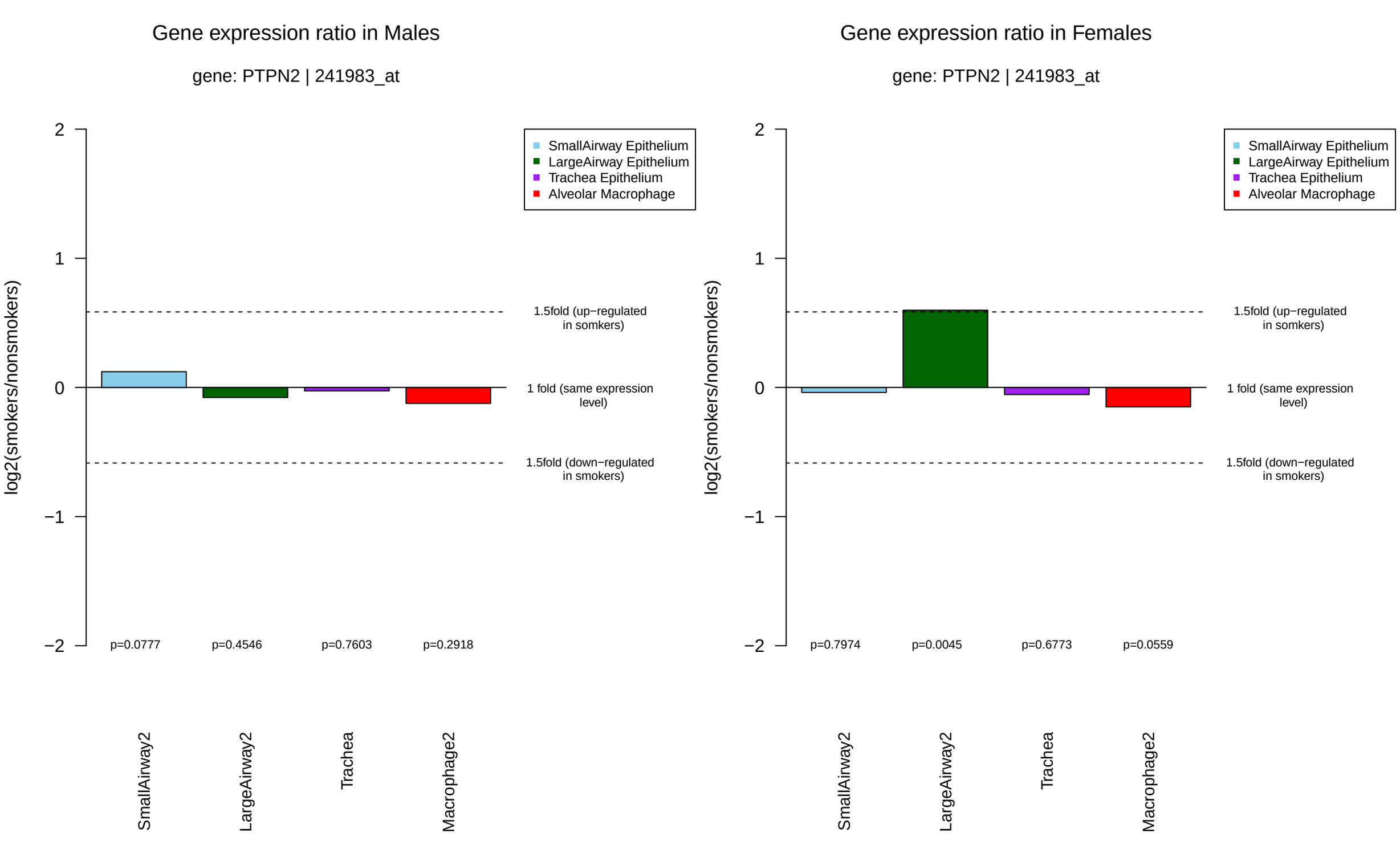
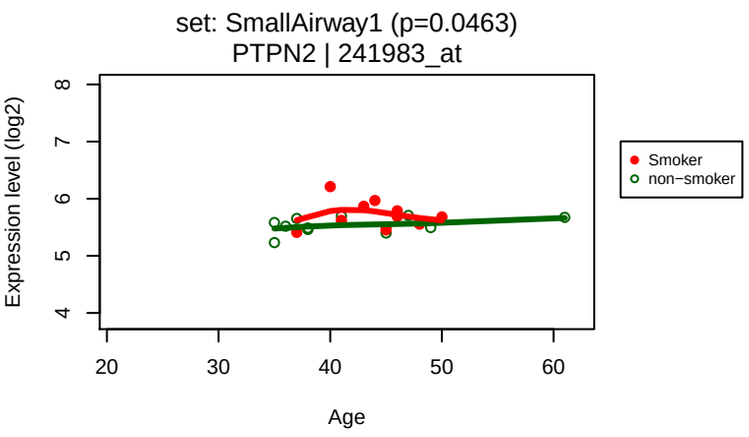
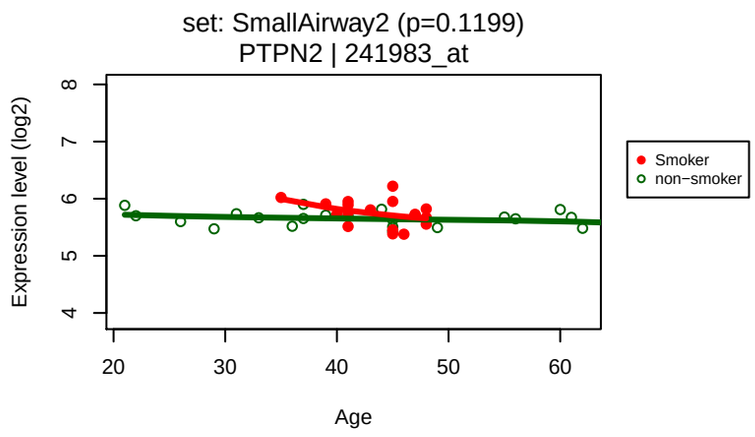
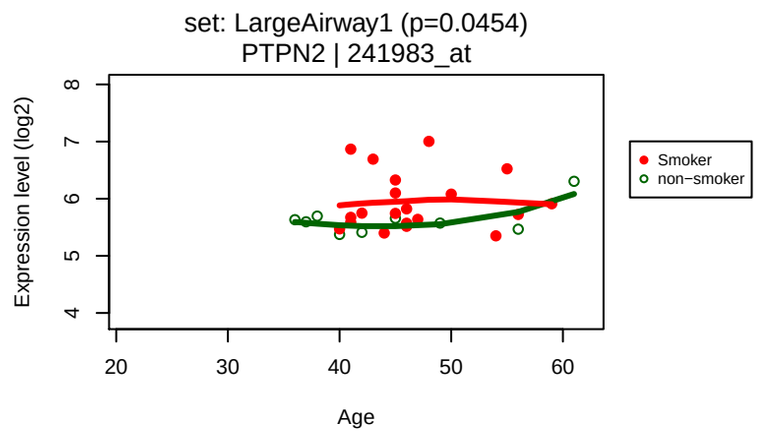

Gene expression of smokers and nonsmokers
gene: PTPN2 | 241983_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.0463
p=0.1199
p=0.2049
p=0.0454
p=0.1301
p=0.7629
p=0.7254
p=0.0993
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

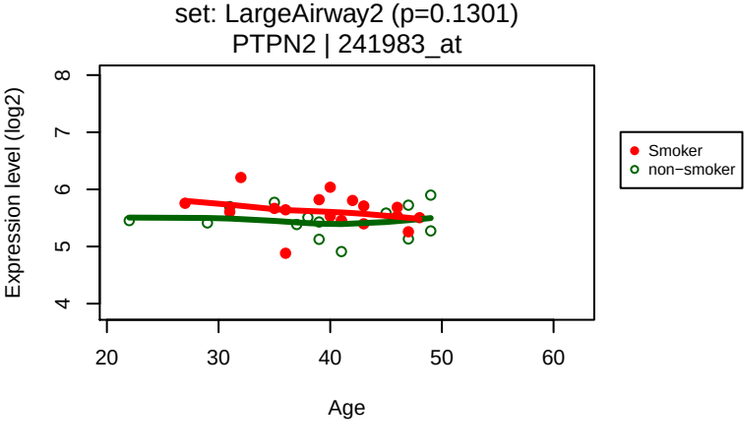

Gene expression ratio (smokers/nonsmokers)
gene: PTPN2 | 241983_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0463
p=0.1199
p=0.2049
p=0.0454
p=0.1301
p=0.7629
p=0.7254
p=0.0993
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

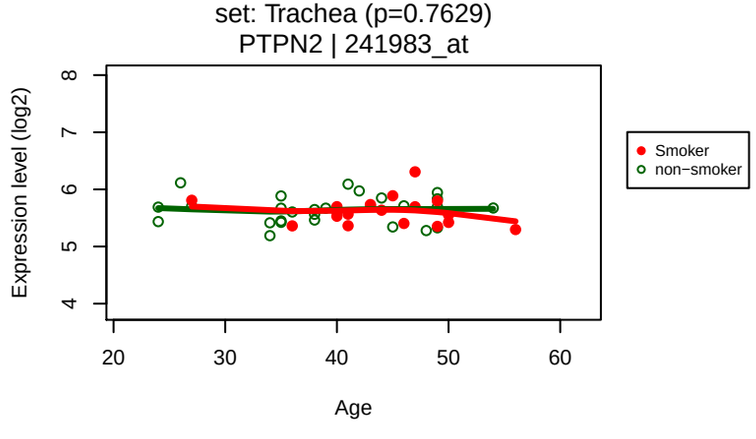

Gene expression ratio in Males
gene: PTPN2 | 241983_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0777
p=0.4546
p=0.7603
p=0.2918
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: PTPN2 | 241983_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.7974
p=0.0045
p=0.6773
p=0.0559
SmallAirway2
LargeAirway2
Trachea
Macrophage2

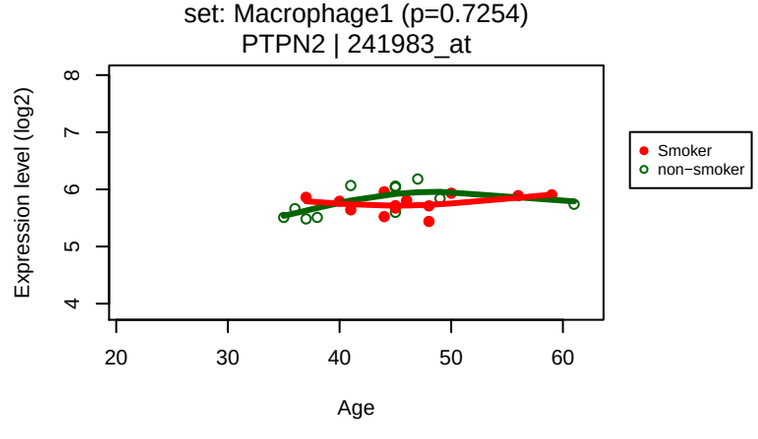

set: SmallAirway1 (p=0.0463)
PTPN2 | 241983_at
Expression level (log2)
Age
Smoker
non−smoker

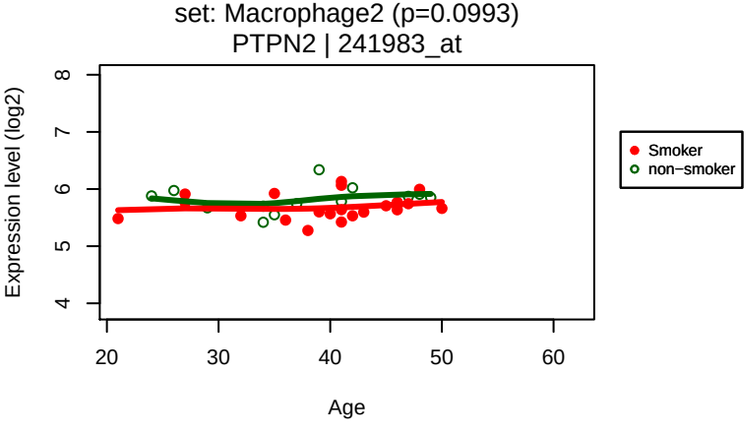

set: SmallAirway2 (p=0.1199)
PTPN2 | 241983_at
Expression level (log2)
Age
Smoker
non−smoker

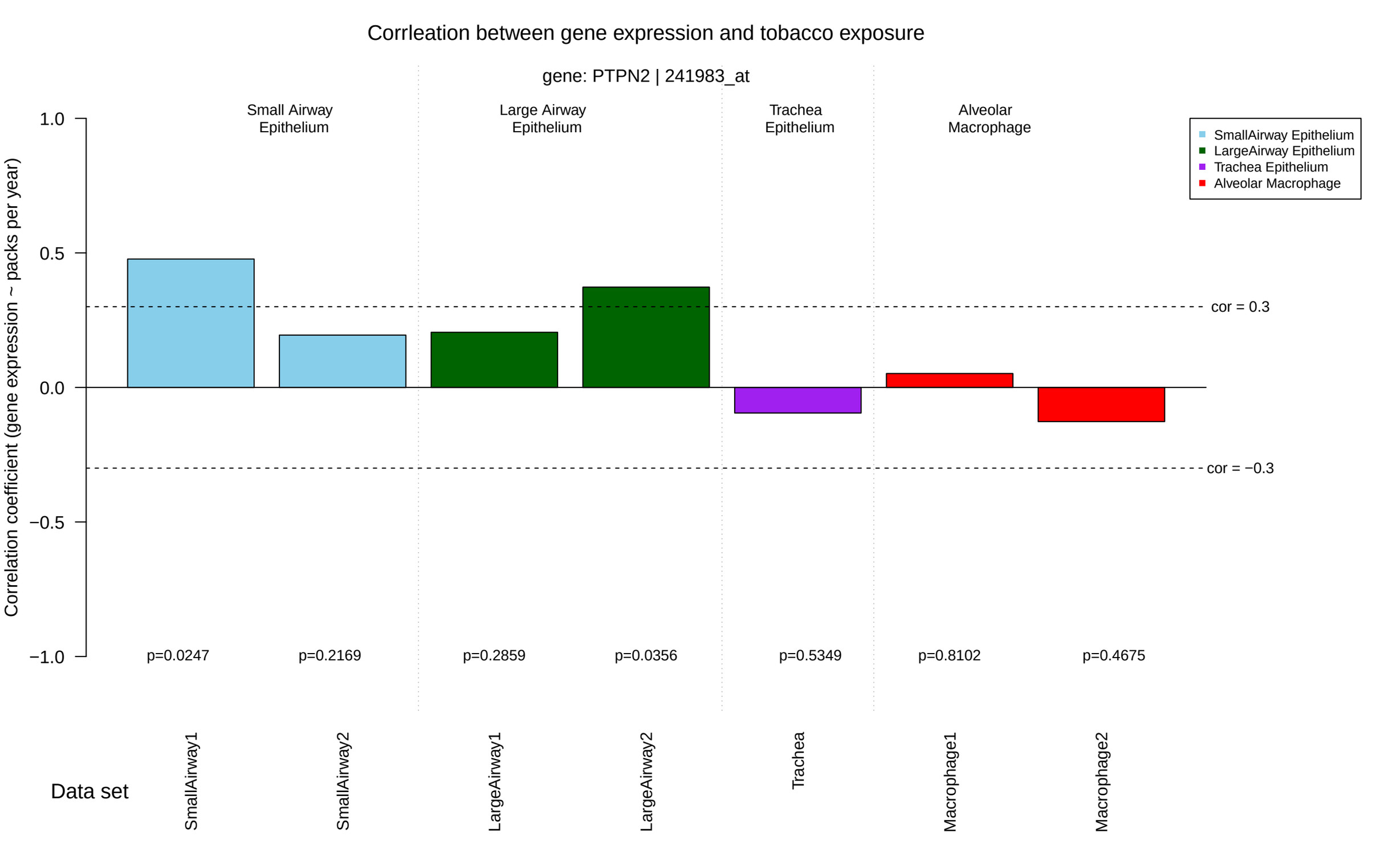

set: LargeAirway1 (p=0.0454)
PTPN2 | 241983_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.1301)
PTPN2 | 241983_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.7629)
PTPN2 | 241983_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.7254)
PTPN2 | 241983_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.0993)
PTPN2 | 241983_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: PTPN2 | 241983_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.0247
p=0.2169
p=0.2859
p=0.0356
p=0.5349
p=0.8102
p=0.4675
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2